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Newton's method for systems of non-linear equations: an application to an ODE problem

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Chapter 1

Introduction

The aim of this thesis is to study the Newton's generalized method for systems of non-linear equations and apply it to mildly non-linear boundary value problems through an implementation of the method in Matlab.

In Chapter 2 we start by considering non-linear equations, we define the iterative methods (for whom we give a convergence's theorem) and the Newton's method for non-linear equations. Then we extend this theory to the multidimensional case: we consider systems of non-linear equations, we study a theorem of local convergence and define the Newton's method for systems of non-linear equations with two convergence's theorems. Lastly we give some variations of the method that are less precise but, if used properly, reduce the computational cost.

In Chapter 3 we study some systems for which the Newton's method is particularly efficient. We start by giving the definition of tridiagonal and diagonally dominant systems and study the theorems for which, under proper assumptions, these systems have one and only one solution. Finally we define the mildly non-linear systems and give a theorem for them to have one and only one solution. Then we show how to apply the Newton's method to this type of systems.

In the last Chapter, Chapter 4, we define the mildly non-linear boundary value problems, we show how to discretize it with the central difference method and consider it as a mildly non-linear system which satisfies the assumption of the theorem of existence and uniqueness of the solution that we have studied in Chapter 3. Now we can use the Newton's generalized method to find such solution. We finish this thesis with some examples.

In Appendix A there is the code, written in Matlab, of the Newton's method.

Chapter 2

Newton's method for systems of nonlinear equations

In this Chapter, after a brief introduction about the theory of iterative methods for the resolution of non-linear equations, we will extend them to the multidimensional case of systems of non-linear equations. In particular we will define the Newton's method, assuming that the systems have one and only one solution., then we will study the convergence's theorems and finally some variations of the Newton's method. In order to study it we have followed Ref. [1].

2.1 Resolution of scalar non-linear equations

We start by considering a system with just one scalar non-linear equation. Given a non-linear function $f : \mathbb{R} \rightarrow \mathbb{R}$, we have to find the scalar $\alpha \in \mathbb{R}$ such that

$$f(\alpha) = 0. \quad (2.1)$$

The value α is called *zero*, *root*, or *solution* of the equation.

The numerical methods that solve this type of problems are called *iterative* methods; their idea is to change the problem from $f(x) = 0$ to the equivalent one $x - \phi(x) = 0$, where the auxiliary function $\phi : \mathbb{R} \rightarrow \mathbb{R}$ has to be chosen such that $\phi(\alpha) = \alpha$ every time that $f(\alpha) = 0$. Given an initial value $x^{(0)}$, we generate a succession of vectors $\{x^{(k)}\}_{k \geq 0}$ defined as:

$$x^{(k+1)} = \phi(x^{(k)}), \quad k \geq 0. \quad (2.2)$$

An iterative method is *convergent* if the succession $\{x^{(k)}\}_{k \geq 0}$ converges to a solution α of the problem, that is if $\lim_{k \rightarrow \infty} x^{(k)} = \alpha$. It is *locally convergent* if the convergence critically depends on the choice of $x^{(0)}$ near α .

A convergent succession $\{x^{(k)}\}_{k \geq 0}$ has a convergence order ρ defined as follows: letting $e_k = x^{(k)} - \alpha$ if there exist two real numbers ρ and $C \neq 0$ such that

$$\lim_{k \rightarrow \infty} \frac{|e_{k+1}|}{|e_k|^\rho} = C,$$

then the succession has order of convergence ρ and factor of convergence C . For $\rho = 1$ the convergence is said linear, for $\rho = 2$ quadratic. When $\rho = 1$ the convergence to α implies $C < 1$.

From the following theorem we can obtain a sufficient condition for the local convergence of the method.

Theorem 2.1.1. *Let α be a fixed point of $\phi(x)$ in an interval J where $\phi(x)$ is derivable with continuity and let's suppose that there exist two positive numbers η and K with $K < 1$ such that $\forall x \in [\alpha - \eta, \alpha + \eta] \subset J$,*

$$|\phi'(x)| \leq K; \quad (2.3)$$

then the following propositions are true:

1. *if $x^{(0)} \in]\alpha - \eta, \alpha - \eta[$, then also $x^{(k)} \in]\alpha - \eta, \alpha - \eta[$ for $k = 1, 2, \dots$;*
2. *for the succession $\{x^{(k)}\}_{k \geq 0}$, with $x^{(0)} \in]\alpha - \eta, \alpha - \eta[$ we have $\lim_{k \rightarrow \infty} x^{(k)} = \alpha$;*
3. *α is the only fixed point of $\phi(x)$ in $[\alpha - \eta, \alpha + \eta]$.*

2.1.1 Newton's method for non-linear equations

Assuming $f \in C^1(J)$, J interval, and $f'(x) \neq 0 \forall x \in J \setminus \{\alpha\}$, the Newton's method is:

$$\begin{aligned} &\text{given } x^{(0)} \\ &x^{(k+1)} = x^{(k)} - \frac{f(x^{(k)})}{f'(x^{(k)})}, \quad k \geq 0, \quad f'(x^{(k)}) \neq 0. \end{aligned} \quad (2.4)$$

At the k -th iteration, this method requires two functional evaluations, $f(x^{(k)})$ and $f'(x^{(k)})$. This increases the computational cost compared with other methods like the bisection method or the Regula Falsi method, but on the other hand it also increases the convergence' speed. Indeed, when α is a simple root ($f'(\alpha) \neq 0$) it converges with order $\rho = 2$, while if α is a multiple root, it locally converges with order $\rho = 1$.

2.2 Resolution of systems of non-linear equations

The theory of iterative methods for non-linear equations can be extended to the multidimensional case, where $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is continuously differentiable. In order to find a fixed point α such that $F(\alpha) = 0$, an appropriate iterative function is

$$\phi(\mathbf{x}) = \mathbf{x} - G(\mathbf{x})F(\mathbf{x}), \quad (2.5)$$

where

$$\begin{aligned} \mathbf{x} &= (x_1, \dots, x_n), \\ \phi(\mathbf{x}) &= (\phi_1(\mathbf{x}), \dots, \phi_n(\mathbf{x})), \\ F(\mathbf{x}) &= (f_1(\mathbf{x}), \dots, f_n(\mathbf{x})), \end{aligned} \quad (2.6)$$

and $G(\mathbf{x})$ is a non singular $n \times n$ matrix in a domain $D \subseteq \mathbb{R}^n$ containing α .

We will then consider methods with

$$\mathbf{x}^{(k+1)} = \phi(\mathbf{x}^{(k)}). \quad (2.7)$$

If there exist the first derivative of every functions ϕ_i , introducing the Jacobian matrix,

$$J_\phi(\mathbf{x}) = \begin{pmatrix} \frac{\partial \phi_1}{\partial x_1} & \cdots & \frac{\partial \phi_1}{\partial x_n} \\ \vdots & & \vdots \\ \frac{\partial \phi_n}{\partial x_1} & \cdots & \frac{\partial \phi_n}{\partial x_n} \end{pmatrix} \quad (2.8)$$

we can generalize the theorem (2.1.1) of the scalar case to the following one:

Theorem 2.2.1 (Theorem of local convergence).

If α is a fixed point of $\phi(\mathbf{x})$, a sufficient condition for the convergence to α of the method is that there exist two positive number K and η , with $K < 1$ such that

$$\|J_\phi(\mathbf{x})\| \leq K, \forall \mathbf{x} \in D_\eta = \{\mathbf{x} \text{ s.t } \|\mathbf{x} - \alpha\| \leq \eta\} \quad (2.9)$$

as long as $\mathbf{x}^{(0)}$ is chosen in D_η . In that case α is the only fixed point of ϕ in D_η .

In addition, if there exist the second derivatives of ϕ_i , we can demonstrate that, if the method converge to a fixed point α , a sufficient condition for the linear convergence is

$$J_\phi(\alpha) \neq 0. \quad (2.10)$$

If $J_\phi(\alpha) = 0$ the convergence is at least quadratic and it is exactly quadratic if it isn't null at least one of the hessian matrices

$$H_i(\alpha) = \begin{pmatrix} \frac{\partial^2 \phi_i}{\partial x_1^2} & \cdots & \frac{\partial^2 \phi_i}{\partial x_1 \partial x_n} \\ \vdots & & \vdots \\ \frac{\partial^2 \phi_i}{\partial x_n \partial x_1} & \cdots & \frac{\partial^2 \phi_i}{\partial x_n^2} \end{pmatrix}. \quad (2.11)$$

2.3 Newton's method for systems of non-linear equations

In order to extend the Newton's method to the non-linear systems, we require that the $f_i(\mathbf{x})$ are derivable with continuity for each variable and that the Jacobian matrix is not singular in a domain containing the system' solution, α . The iteration function takes the form

$$\phi(\mathbf{x}) = \mathbf{x} - J_F^{-1}(\mathbf{x})F(\mathbf{x}), \quad (2.12)$$

so the system is

$$\mathbf{x} = \mathbf{x} - J_F^{-1}(\mathbf{x})F(\mathbf{x}),$$

from which we can derive the Newton's (or Newton-Raphson) iterative method:

$$\begin{aligned} &\text{given } \mathbf{x}^{(0)} \in \mathbb{R}^n \\ &\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} - J_F^{-1}(\mathbf{x}^{(k)})F(\mathbf{x}^{(k)}), \quad k = 0, 1, \dots, \quad J_F(\mathbf{x}^{(k)}) \text{ non singular.} \end{aligned} \quad (2.13)$$

In the practice the iterate $\mathbf{x}^{(k+1)}$ is obtained from the solution of the linear system

$$J_F(\mathbf{x}^{(k)})\mathbf{d}^{(k)} = -F(\mathbf{x}^{(k)}), \quad k = 0, 1, \dots \quad (2.14)$$

where $\mathbf{d}^{(k)} = \mathbf{x}^{(k+1)} - \mathbf{x}^{(k)}$.

For every step k we have to solve a linear system with matrix $J_F(\mathbf{x}^{(k)})$.

This method is extremely sensible to the choice of $\mathbf{x}^{(0)}$, so the convergence is only locally guaranteed.

2.4 Convergence's theorems

Let's see now two theorems which guarantee us the local convergence of the Newton's method under different hypothesis.

Theorem 2.4.1. *Let $\alpha \in D$ be the solution of $F(\mathbf{x}) = 0$, with the functions $f_i(\mathbf{x})$ of class $C^3(D)$ and such that $J_F(\mathbf{x})$ is not singular in D ; then the order of convergence of the Newton's method is at least $\rho = 2$ and there is convergence for every choice of $\mathbf{x}^{(0)}$ in a proper domain containing α .*

Proof. Considering the form of the function $\phi(\mathbf{x})$, the j -th column of the matrix is

$$\begin{aligned} \frac{\partial \phi(\mathbf{x})}{\partial x_j} &= \frac{\partial \mathbf{x}}{\partial x_j} - \frac{\partial}{\partial x_j} [J_F^{-1}(\mathbf{x})f(\mathbf{x})] \\ &= \frac{\partial \mathbf{x}}{\partial x_j} - J_F^{-1}(\mathbf{x}) \frac{\partial f(\mathbf{x})}{\partial x_j} - \frac{\partial J_F^{-1}(\mathbf{x})}{\partial x_j} f(\mathbf{x}) \\ &= -\frac{\partial J_F^{-1}(\mathbf{x})}{\partial x_j} f(\mathbf{x}); \end{aligned}$$

as the vectors $\frac{\partial \mathbf{x}}{\partial x_j}$ and $J_F^{-1}(\mathbf{x}) \frac{\partial f(\mathbf{x})}{\partial x_j}$ coincide with the j -th column of the identical matrix. It follows that for $\mathbf{x} = \alpha$ there is

$$\left(\frac{\partial \phi(\mathbf{x})}{\partial x_j} \right)_{\mathbf{x}=\alpha} = - \left(\frac{\partial J_F^{-1}(\mathbf{x})}{\partial x_j} \right)_{\mathbf{x}=\alpha} f(\alpha) = 0, \quad j = 1, 2, \dots, n.$$

Thus,

$$J_\phi(\alpha) = 0$$

meaning that the convergence's order of the method is at least $\rho = 2$. From the continuity of $\phi(x)$ and from $J_\phi(\alpha) = 0$ follows that, for a chosen $K < 1$, there exist a domain $D_\eta = \{\mathbf{x} \text{ s.t. } \|\mathbf{x} - \alpha\| \leq \eta\} \subset D$ in which it's valid the condition (4.33) and so the method converges $\forall \mathbf{x}^{(0)} \in D_\eta$. $\square$

Theorem 2.4.2. *Let $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a vectorial function of class C^1 on a convex open set D of $\mathbb{R}^n$ containing α . Let's suppose that $J_F^{-1}(\alpha)$ exists and that there exist three positive constants R, C and L such that $\|J_F^{-1}(\alpha)\| \leq C$ and*

$$\|J_F(\mathbf{x}) - J_F(\mathbf{y})\| \leq L\|\mathbf{x} - \mathbf{y}\| \quad \forall \mathbf{x}, \mathbf{y} \in B(\alpha; R),$$

where with the same symbol $\|\cdot\|$ we mean a vectorial norm and a consistent matricial norm. Then there exist an $r > 0$ such that, for every $\mathbf{x}^{(0)} \in B(\alpha, R)$, the succession of the Newton's method is uniquely defined, converges to α and

$$\|\mathbf{x}^{(k+1)} - \alpha\| \leq CL\|\mathbf{x}^{(k)} - \alpha\|^2. \quad (2.15)$$

Proof. Proceeding by induction on k , we verify that (2.15) is true, and also that $\mathbf{x}^{(k+1)} \in B(\alpha, r)$, where $r = \min(R, 1/(2CL))$. Let's prove, above all, that for every $\mathbf{x}^{(0)} \in B(\alpha, r)$, the matrix $J_F^{-1}(\mathbf{x}^{(0)})$ exists. Indeed, we have

$$\|J_F^{-1}(\alpha)[J_F(\mathbf{x}^{(0)}) - J_F(\alpha)]\| \leq \|J_F^{-1}(\alpha)\| \|\alpha - \mathbf{x}^{(0)}\| \leq CLr \leq \frac{1}{2}$$

and then, we can conclude that $J_F^{-1}(\mathbf{x}^{(0)})$ exists since

$$J_F^{-1}(\mathbf{x}^{(0)}) \leq \frac{\|J_F^{-1}(\alpha)\|}{1 - \|J_F^{-1}(\alpha)[J_F(\mathbf{x}^{(0)}) - J_F(\alpha)]\|} \leq 2\|J_F^{-1}(\alpha)\| \leq 2C.$$

As a result, $\mathbf{x}^{(1)}$ is well defined and

$$\mathbf{x}^{(1)} - \mathbf{x}^{(0)} = \mathbf{x}^{(0)} - \alpha - J_F^{-1}(\mathbf{x}^{(0)})[F(\mathbf{x}^{(0)}) - F(\alpha)].$$

Grouping at the second member $J_F^{-1}(\mathbf{x}^{(0)})$ and moving to the norms, we obtain

$$\|\mathbf{x}^{(1)} - \alpha\| \leq \|J_F^{-1}(\mathbf{x}^{(0)})\| \|F(\alpha) - F(\mathbf{x}^{(0)}) - J_F(\mathbf{x}^{(0)})[\alpha - \mathbf{x}^{(0)}]\| \leq 2C\frac{L}{2}\|\alpha - \mathbf{x}^{(0)}\|^2$$

by using the remainder of the Taylor expansion of F . The previous relation proves the (2.15) when $k = 0$. In addition, since $\mathbf{x}^{(0)} \in B(\alpha, r)$ we have $\|\alpha - \mathbf{x}^{(0)}\| \leq 1/(2CL)$, from which $\|\mathbf{x}^{(1)} - \alpha\| \leq \|\alpha - \mathbf{x}^{(0)}\|/2$. This ensures that $\mathbf{x}^{(1)} \in B(\alpha, r)$. With a similar procedure we can prove that if the (2.15) applies for a certain k , then it has to be true also for $k + 1$. This proves the theorem. $\square$

In conclusion the method converges with order 2 if $\mathbf{x}^{(0)}$ is close enough to α and $J_F(\mathbf{x}^{(k)})$ is not singular, $\forall k \geq 0$

If α is a multiple root, thus $J_F(\alpha)$ is singular and the method converges with order 1.

2.5 Variations of the Newton's method

We introduce now some variations of the Newton's method in order to reduce its computational cost if $\mathbf{x}^{(0)}$ is close enough to α .

2.5.1 Newton's method simplified

If there is a slow variation of the Jacobian matrix $J_F(\mathbf{x})$ we can use the simplified Newton's method, which is

$$J_F(\mathbf{x}^{(0)})\mathbf{d}^{(k)} = -f(\mathbf{x}^{(k)}), \quad k = 0, 1, \dots, \quad (2.16)$$

where the Jacobian matrix is evaluated only once, in the correspondence of a good approximation $\mathbf{x}^{(0)}$. If the method converges, the convergence is usually linear.

Likewise we can keep the Jacobian matrix fixed for a certain number of $p \geq 2$ steps of the method. In general, to the rise of the efficiency corresponds a performance decline in terms of speed of convergence.

2.5.2 Jacobi-Newton's non-linear method

A way to avoid the calculation of the entire matrix $J_F(\mathbf{x})$ at every step is to consider the i -th equation of the system (2.6) as a single equation in the unknown x_i and to apply to every equation of the system the Newton's method for equations in one variable. If we assume that the sorting of the equations is such that, in a domain containing α , there is

$$\frac{\partial f_i(x_1, \dots, x_n)}{\partial x_i} \neq 0 \quad i = 1, 2, \dots, n \quad (2.17)$$

then we obtain

$$x_i^{(k+1)} = x_i^{(k)} - \frac{f_i(x_1^{(k)}, \dots, x_n^{(k)})}{\frac{\partial f_i(x_1^{(k)}, \dots, x_n^{(k)})}{\partial x_i}} \quad i = 1, \dots, n; \quad k = 0, 1, \dots \quad (2.18)$$

This is a one point iterative method and it can be expressed in the matricial form

$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} - D^{-1}(\mathbf{x}^{(k)})f(\mathbf{x}^{(k)}), \quad k = 0, 1, \dots, \quad (2.19)$$

where $D = \text{diag}(\frac{\partial f_1}{\partial x_1}, \frac{\partial f_2}{\partial x_2}, \dots, \frac{\partial f_n}{\partial x_n})$. The convergence is linear and it's granted if it's true the condition (2.9), considering that in this method there is $\phi(\mathbf{x}) = \mathbf{x} - D^{-1}(\mathbf{x})f(\mathbf{x})$.

2.5.3 Gauss-Seidel non-linear method

This variation is obtained by introducing the Gauss-Seidel's iterative technique:

$$x_i^{(k+1)} = x_i^{(k)} - \frac{f_i(x_1^{(k+1)}, \dots, x_{i-1}^{(k+1)}, x_i^{(k)}, \dots, x_n^{(k)})}{\frac{\partial f_i(x_1^{(k+1)}, \dots, x_{i-1}^{(k+1)}, x_i^{(k)}, \dots, x_n^{(k)})}{\partial x_i}} \quad i = 1, \dots, n; \quad k = 0, 1, \dots \quad (2.20)$$

where for the calculation of $x_i^{(k+1)}$ we have used the first $i - 1$ components already known of the iterate $\mathbf{x}^{(k+1)}$ in the progress of the calculation.

2.5.4 Newton's Over Relaxation method

A way to accelerate the linear convergence of this method is to introduce a relaxation factor $\omega \in]0, 2[$, setting

$$x_i^{(k+1)} = x_i^{(k)} - \omega F_i^{(k+1,k)} \quad i = 1, 2, \dots, n, \quad k = 0, 1, \dots \quad (2.21)$$

where $F_i^{(k+1,k)}$ is the fraction that appears in (2.20). The (2.21) with $\omega > 1$ is called the Newton's Over Relaxation method.

2.5.5 Other methods

There is another family of methods, derived from the Newton-Raphson's method, that can be interpreted as an extension of the secant method in $\mathbb{R}^n$. They are obtained by (2.14) replacing $J_f(\mathbf{x})$ with one of its discrete approximations; in this way we have the advantage of not having to calculate the n^2 partial derivatives in $J_f(\mathbf{x})$. In order to approximate $J_f(\mathbf{x}^{(k)})$ we can introduce the matrix $\Delta(\mathbf{x}^{(k)})$, real and of order n , whose elements are given by

$$(\Delta(\mathbf{x}^{(k)}))_{ij} = \frac{f_i(\mathbf{x}^{(k)} + h_j^{(k)} \mathbf{e}^{(j)}) - f_i(\mathbf{x}^{(k)})}{h_j^{(k)}} \quad i, j = 1, 2, \dots, n \quad k = 0, 1, \dots, \quad (2.22)$$

where $\mathbf{e}^{(j)}$ is the j -th vector of the canonical basis and $h_j^{(k)}$ are appropriate scalars, infinitesimals at the growth of k ; for example we can assume $h_j^{(k)} = f_j(\mathbf{x}^{(k)})$, or, easier, $h_j^{(k)} = x_j^{(k-1)} - x_j^{(k)}$. In general, this methods have a convergence order of $\rho \leq 2$.

Chapter 3

Mildly non-linear systems

The Newton's method is particularly efficient when the system that has to be solved is mildly non-linear, tridiagonal and diagonally dominant, so in this chapter we will give the definition of these properties and the result of existence and uniqueness for such systems following ref. [1].

3.1 Tridiagonal and Diagonally dominant Systems

The general linear system of n equations in n unknown $x_1, x_2, x_3, \dots, x_n$ can be written in the following form:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \dots + a_{2n}x_n &= b_2 \\ \cdot & \quad \cdot \quad \cdot \quad \cdot \quad \cdot \quad \cdot \quad \cdot \\ \cdot & \quad \cdot \quad \cdot \quad \cdot \quad \cdot \quad \cdot \quad \cdot \\ a_{n1}x_1 + a_{n2}x_2 + a_{n3}x_3 + \dots + a_{nn}x_n &= b_n, \end{aligned} \tag{3.1}$$

or, more compactly as

$$\mathbf{Ax} = \mathbf{b}, \tag{3.2}$$

where

$$A = \begin{bmatrix} a_{11} & a_{12} & 0 & \cdot & \cdot & \cdot & 0 \\ a_{21} & a_{22} & a_{23} & 0 & \cdot & \cdot & 0 \\ 0 & a_{32} & a_{33} & a_{34} & 0 & \cdot & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ 0 & \cdot & \cdot & \cdot & 0 & a_{n,n-1} & a_{nn} \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \cdot \\ \cdot \\ \cdot \\ x_n \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ \cdot \\ \cdot \\ \cdot \\ b_n \end{bmatrix}. \tag{3.3}$$

The following theorem assures the existence and uniqueness of the system' solution.

Theorem 3.1.1. *A system like (3.1) has one and only one solution if and only if the determinant of A is different from zero.*

System (3.1) is said to be *diagonally dominant* if and only if

$$|a_{ii}| \geq \sum_{j=1, j \neq i}^n |a_{ij}|, \quad i = 1, 2, \dots, n, \quad (3.4)$$

with strict inequality valid for at least one value of i .

System (3.1) is said to be *tridiagonal* if and only if all elements of A are zero except a_{ii} , $a_{j,j+1}$, $a_{j+1,j}$, $i = 1, 2, \dots, n$, $j = 1, 2, \dots, n-1$.

In the matrix form all the elements of a tridiagonal system are zero except those on the three diagonals: the main diagonal, the superdiagonal and the subdiagonal.

There are certain tridiagonal and diagonally dominant systems in which the main diagonal elements are negative and the subdiagonal and the superdiagonal elements are positive. We will show that for such systems $|A| \neq 0$, and, as a result, that they have one and only one solution. In anticipation of that we give two preliminary results.

Lemma 3.1.2. *If $\mathbf{x} = a_1x_1 + a_2x_2 + \dots + a_kx_k$ for k positive numbers $a_1, \dots, a_k$ which satisfy $a_1 + a_2 + \dots + a_k = 1$, then*

$$\min_i(x_i) \leq \mathbf{x} \leq \max_i(x_i), \quad (3.5)$$

where the equal signs apply if and only if $x_1 = x_2 = \dots = x_k$.

Lemma 3.1.3. *If $\mathbf{x} = a_1x_1 + a_2x_2 + \dots + a_kx_k$ for k positive numbers $a_1, \dots, a_k$ which satisfy $a_1 + a_2 + \dots + a_k < 1$, then*

$$\min(0, x_1, \dots, x_k) \leq \mathbf{x} \leq \max(0, x_1, \dots, x_k), \quad (3.6)$$

where the equal signs apply if and only if $x_1 = x_2 = \dots = x_k = 0$.

Now we can state a major theorem.

Theorem 3.1.4. *If a system is tridiagonal, diagonally dominant, and satisfy*

$$\begin{aligned} a_{ii} &< 0 & i &= 1, 2, \dots, n, \\ a_{j,j+1} &> 0 & j &= 1, 2, \dots, n-1, \\ a_{j+1,j} &> 0 & j &= 1, 2, \dots, n-1, \end{aligned} \quad (3.7)$$

then, the system has one and only one solution.

Proof. We only need to show that $|A| \neq 0$, and then we can use Theorem (3.1.1). This can be done showing that the only solution of the related homogeneous system

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 &= 0 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 &= 0 \\ a_{32}x_2 + a_{33}x_3 + a_{34}x_4 &= 0 \\ \cdot &\quad \cdot &\quad \cdot &\quad \cdot &\quad \cdot &\quad \cdot \\ a_{n,n-1}x_{n-1} + a_{nn}x_n &= 0 \end{aligned} \quad (3.8)$$

is $x_1 = x_2 = \dots = x_n = 0$. In order to do that we assume that the (3.8) has a nonzero solution and show that this must lead to a contradiction. Let's assume that (3.8) has a solution with at least one of $x_1, x_2, \dots, x_n$ not zero. Without losing generality, we can assume that one of these values is positive (it is analogue if we assume it negative). If one between $x_1, x_2, \dots, x_n$ is positive, then there is a maximum positive number in this set, since the numbers in consideration are n . Let x_k satisfy

$$x_k \geq x_i, \quad i = 1, 2, \dots, n, \quad (3.9)$$

and let's consider the three possibilities: $k = 1, k = n$ and $1 < k < n$.

If $k = 1$, then (3.9) becomes

$$x_1 \geq x_i, \quad i = 1, 2, \dots, n. \quad (3.10)$$

But then the first equation in (3.8) implies

$$x_2 = -\frac{a_{11}}{a_{12}}x_1. \quad (3.11)$$

However, $-(a_{11}/a_{12}) > 0$ by (3.7), so that $x_2 > 0$. But, by diagonal dominance, we have $-(a_{11}/a_{12}) \geq 1$, so that $x_2 \geq x_1$. However, $x_1 \geq x_2$, so that $x_1 = x_2$ is also maximal and positive and

$$-a_{11} = a_{12}. \quad (3.12)$$

For the case $k = n$ a completely analogous argument yields $x_{n-1} = x_n$ and

$$-a_{nn} = a_{n,n-1}. \quad (3.13)$$

Consider now the case $1 < k < n$. Then

$$x_k \geq x_i, \quad i = 1, 2, \dots, n. \quad (3.14)$$

But the k -th equation in (3.8) implies

$$x_k = -\frac{a_{k,k-1}}{a_{kk}}x_{k-1} - \frac{a_{k,k+1}}{a_{kk}}x_{k+1}. \quad (3.15)$$

However, by (3.7) we have $-(a_{k,k-1}/a_{kk}) > 0$ and $-(a_{k,k+1}/a_{kk}) > 0$. Moreover, by diagonal dominance,

$$-\frac{a_{k,k-1}}{a_{kk}} - \frac{a_{k,k+1}}{a_{kk}} \leq 1. \quad (3.16)$$

Strict inequality cannot apply in (3.16) since, otherwise, Lemma (3.1.3), applied to (3.15), would imply that $x_k \leq \max(0, x_{k-1}, x_{k+1})$ where the equal sign applies only if $x_{k-1} = x_{k+1}$. But this is not possible since $x_k = \max(x_1, x_2, \dots, x_n) > 0$. Consequently we have

$$-a_{kk} = a_{k,k-1} + a_{k,k+1}. \quad (3.17)$$

Moreover, Lemma (3.1.2) applied to (3.15) yields $x_k \leq \max(x_{k-1}, x_{k+1})$, and since x_k is maximal and positive by hypothesis, we have

$$x_k = x_{k-1} = x_{k+1}. \quad (3.18)$$

Thus, x_{k-1} and x_{k+1} are also maximal and positive. Consequently, the above analysis can be repeated for x_j , $j = k - 1, k - 2, \dots, 2$, and $j = k + 1, k + 2, \dots, n - 1$, so that

$$-a_{jj} = a_{j,j-1} + a_{j,j+1}, \quad j = 2, 3, \dots, n - 1. \quad (3.19)$$

From the above discussion, it follows that $x_1 = x_2 = \dots = x_n$. Thus, (3.12), (3.13) and (3.19) are valid, which implies

$$-a_{11} = a_{12}, \quad -a_{ii} = a_{i,i-1} + a_{i,i+1}, \quad i = 2, \dots, n - 1, \quad -a_{nn} = a_{n,n-1}. \quad (3.20)$$

But $a_{ii} < 0$ implies $-a_{ii} = |a_{ii}|$, while $a_{ij} \geq 0, i \neq j$ implies $a_{ij} = |a_{ij}|$. Thus, (3.20) can be written as

$$|a_{11}| = |a_{12}|, \quad |a_{ii}| = |a_{i,i-1}| + |a_{i,i+1}|, \quad i = 2, \dots, n - 1, \quad |a_{nn}| = |a_{n,n-1}|. \quad (3.21)$$

Finally, we note that (3.21) contradicts the assumption of diagonal dominance, since strict inequality must be valid for at least one $i, i = 1, 2, \dots, n$. Thus, by contradiction, the system (3.8) has only the zero solution and $|A| \neq 0$. $\square$

3.2 Mildly non-linear systems

A *mildly non-linear system* is a system that has the following form:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n + f_1(x_1) &= 0 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n + f_2(x_2) &= 0 \\ \cdot &\cdot \cdot \cdot \cdot \cdot \cdot \cdot \\ a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nn}x_n + f_n(x_n) &= 0, \end{aligned} \quad (3.22)$$

where $f_i : \mathbb{R} \rightarrow \mathbb{R}$, $f_i(x_i)$, $i = 1, 2, \dots, n$ are non-linear functions in only one variable.

The following theorem guarantees the existence and uniqueness of the solution of particular mildly non-linear systems.

Theorem 3.2.1. *Let the linear part of a mildly non linear system be tridiagonal, diagonally dominant and such that*

$$\begin{aligned} a_{ii} &< 0, & i &= 1, 2, \dots, n, \\ a_{j,j+1} &> 0, & j &= 1, 2, \dots, n - 1, \\ a_{j+1,j} &> 0, & j &= 1, 2, \dots, n - 1. \end{aligned} \quad (3.23)$$

If $f_i(x_i)$, $i = 1, 2, \dots, n$, are differentiable and $f'_i(x_i) \leq 0$ for all $\mathbf{x}$, then the system has one and only one solution.

Proof. Let's assume that $x_1, x_2, \dots, x_n$ and $y_1, y_2, \dots, y_n$ are two solutions of (4.7). Then,

$$\sum_{j=1}^n a_{ij}x_j + f_i(x_i) = 0, \quad i = 1, 2, \dots, n, \quad (3.24)$$

and

$$\sum_{j=1}^n a_{ij}y_j + f_i(y_i) = 0, \quad i = 1, 2, \dots, n. \quad (3.25)$$

Then, by subtracting (3.24) from (3.25), we obtain

$$\sum_{j=1}^n a_{ij}(y_j - x_j) + [f_i(y_i) - f_i(x_i)] = 0 \quad i = 1, 2, \dots, n. \quad (3.26)$$

Since $f_i(x_i)$ is differentiable we have, by the Mean Value Theorem,

$$f_i(y_i) - f_i(x_i) = f'_i(\xi_i)(y_i - x_i), \quad (3.27)$$

where ξ_i lies between x_i and y_i . Setting $e_i = y_i - x_i$, the (3.26) become

$$\sum_{j=1}^n a_{ij}e_j + f'_i(\xi_i)e_i = 0, \quad i = 1, 2, \dots, n. \quad (3.28)$$

Now, the (3.28) is a linear system in the unknowns $e_1, e_2, \dots, e_n$. Since $f'_i(\xi_i) \leq 0$, this system satisfies the conditions of Theorem (3.1.3), and hence it has a unique solution that is $e_1 = e_2 = \dots = e_n = 0$. Thus, $x_1 = y_1, x_2 = y_2, \dots, x_n = y_n$. $\square$

3.2.1 Newton's method for mildly non-linear systems

For mildly non-linear system we can use the Newton's over relaxation method and the formula (2.21) reduce to

$$\begin{aligned} x_1^{(k+1)} &= x_1^{(k)} - \omega \frac{a_{11}x_1^{(k)} + a_{12}x_2^{(k)} + \dots + a_{1n}x_n^{(k)} + f_1(x_1^{(k)})}{a_{11} + f'_1(x_1^{(k)})} \\ x_2^{(k+1)} &= x_2^{(k)} - \omega \frac{a_{21}x_1^{(k+1)} + a_{22}x_2^{(k)} + \dots + a_{2n}x_n^{(k)} + f_2(x_2^{(k)})}{a_{22} + f'_2(x_2^{(k)})} \\ &\vdots \\ x_n^{(k+1)} &= x_n^{(k)} - \omega \frac{a_{n1}x_1^{(k+1)} + a_{n2}x_2^{(k+1)} + \dots + a_{nn}x_n^{(k)} + f_n(x_n^{(k)})}{a_{nn} + f'_n(x_n^{(k)})}. \end{aligned} \quad (3.29)$$

In general it is difficult to determine ω *a priori* but when the system that has to be solved is mildly non-linear and satisfies the assumptions of Theorem (3.2.1) the Newton's over relaxation method converges for all the initial guesses $\mathbf{x}^{(0)}$ and all ω in a subrange $0 < \omega < 2$.

Chapter 4

Application to mildly nonlinear boundary value problems for ODE

In this Chapter we will define the mildly non-linear boundary value problem and then, following ref. [2], we will discretize it with the central difference method. Now we can consider it as a mildly non-linear system which has one and only one solution, as we have seen in Chapter 3. Lastly we can find its solution using the Newton' system we have defined in Chapter 2.

4.1 Mildly non-linear boundary value problems

A mildly non-linear boundary value problem is about finding a continuous function $y(x)$ on $[a, b]$ which has continuous first and second derivatives on $a < x < b$ and which satisfies

$$\begin{aligned} y'' + P(x)y' + F(x, y) &= 0, & a < x < b, \\ y(a) &= \alpha, & y(b) = \beta. \end{aligned} \quad (4.1)$$

In order to assure the existence and uniqueness of the analytical solution of this problem are sufficient these two conditions:

1) we assume that $P(x)$ is continuous on $[a, b]$, so there exists a constant M such that

$$|P(x)| \leq M, \quad a \leq x \leq b; \quad (4.2)$$

2) we assume that $F(x, y)$ is continuous on $[a, b] \times \mathbb{R}$ and has continuous partial derivatives with respect to y which satisfy

$$\frac{\partial F(x, y)}{\partial y} \leq 0, \quad a \leq x \leq b, \quad -\infty \leq y \leq +\infty. \quad (4.3)$$

To solve the problem (4.1) at first we choose $n \in \mathbb{N}$ and divide the interval $[a, b]$ into n equal parts of length $h = (b - a)/n$ by the R_{n+1} set $x_i = a + ih$, $i = 0, 1, 2, \dots, n$. Then at each interior grid point x_i , $i = 1, 2, \dots, n - 1$, we substitute $y''(x_i)$ and $y'(x_i)$ with the central difference formula and this yields to a difference equation approximation of the differential equation:

$$\frac{y_{i+1} - 2y_i + y_{i-1}}{h^2} + P(x_i) \frac{y_{i+1} - y_{i-1}}{2h} + F(x_i, y_i) = 0, \quad (4.4)$$

where $y_i = y(x_i)$, or equivalently, setting $P_i = P(x_i)$, $i = 1, \dots, n-1$

$$[2 - hP_i]y_{i-1} - 4y_i + [2 + hP_i]y_{i+1} + 2h^2F(x_i, y_i) = 0. \quad (4.5)$$

Now, we set

$$\begin{aligned} f_1(y_1) &= 2h^2F(x_1, y_1) + (2 - hP_1)y_0, \\ f_i(y_i) &= 2h^2F(x_i, y_i), \quad i = 2, 3, \dots, n-2, \\ f_{n-1}(y_{n-1}) &= 2h^2F(x_{n-1}, y_{n-1}) + (2 + hP_{n-1})y_n. \end{aligned} \quad (4.6)$$

Rewriting (4.5) for each $i = 1, 2, \dots, n-1$, and inserting the known values in (4.1), we obtain the following mildly non-linear system of $n-1$ equations in the $n-1$ unknowns $y_1, \dots, y_{n-1}$:

$$\begin{aligned} -4y_1 + (2 + hP_1)y_2 &+ f_1(y_1) = 0 \\ (2 - hP_2)y_1 - 4y_2 + (2 + hP_2)y_3 &+ f_2(y_2) = 0 \\ \cdot &\cdot \cdot \cdot \cdot \cdot \cdot \\ (2 - hP_{n-1})y_{n-2} - 4y_{n-1} + f_{n-1}(y_{n-1}) &= 0. \end{aligned} \quad (4.7)$$

Generally, systems like (4.7) may not have a solution or may have not only one. However, if h satisfies the condition

$$Mh < 2, \quad (4.8)$$

then the system (4.7) satisfies the assumptions of the Theorem (3.2.1) and hence it has a unique solution. This solution can be found by the generalized Newton's method.

For those boundary value problems in which M is so large that the condition (4.8) is too restrictive, we can use the upwind difference approximation for $y'(x_i)$. In this case the difference approximation for (4.1) at x_i , is

$$\begin{aligned} \frac{y_{i+1} - 2y_i + y_{i-1}}{h^2} + P_i \frac{y_{i+1} - y_i}{h} + F(x_i, y_i) &= 0, \quad \text{if } P_i \geq 0, \\ \frac{y_{i+1} - 2y_i + y_{i-1}}{h^2} + P_i \frac{y_i - y_{i-1}}{h} + F(x_i, y_i) &= 0, \quad \text{if } P_i < 0, \end{aligned} \quad (4.9)$$

or, more compactly,

$$\frac{y_{i+1} - 2y_i + y_{i-1}}{h^2} + \frac{(|P_i| + P_i)y_{i+1} - 2|P_i|y_i + (|P_i| - P_i)y_{i-1}}{2h} + F(x_i, y_i) = 0, \quad i = 1, 2, \dots, n-1. \quad (4.10)$$

Equations (4.10) constitute a mildly non-linear system which satisfies all the assumptions of Theorem (3.2.1) for any given h . Thus, a unique numerical solution exists and can be determined with the generalized Newton's method.

Of course if $P(x) = 0$, the central difference method and the upwind difference method are the same.

4.2 Examples

Let's consider the mildly non-linear boundary value problem

$$\begin{aligned} y'' - 10y' - e^y + \frac{1}{x^2} + \frac{10}{x} + x &= 0 \\ y(1) &= 0, \quad y(e) = 1. \end{aligned} \quad (4.11)$$

In this case $P(x) = -10$ and $F(x, y) = -e^y + 1/x^2 + 10/x + x$, so that $(\partial F / \partial y) = -e^y < 0$ and $P(x) = -10$. Thus the solution exist and it is unique. The analytical solution of this problem is $y = \log x$, as can be easily verified by substituting it in (4.11).

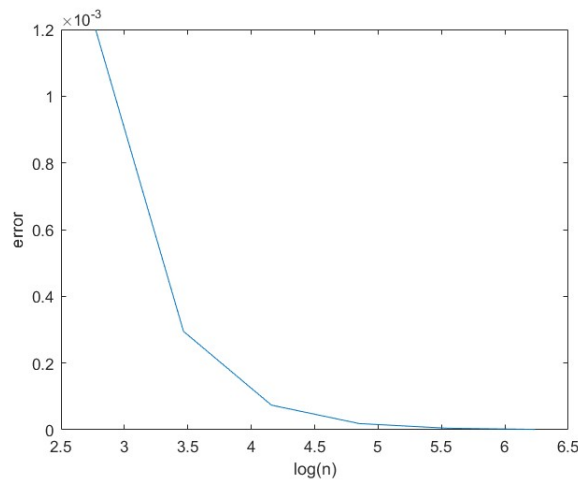
We start by choosing an $n \in \mathbb{N}$, in particular we will use $n = 2^k$, with $k \in \mathbb{N}$, and then divide the interval $[1, e]$ into n parts of length $h = (e - 1)/n$. In this way we have created the vector $\mathbf{x}^{(0)}$ that we will use as initial guess for the method. For $n = 2, 4$, and 8 the condition (4.8) is not satisfied, so it is not ensured that the solution can be found by the Newton's generalized method, instead for $n \geq 16$ the condition is satisfied so we can use the method and analyze the results.

In particular we want to see if the error decreases when n increases. By error we mean the infinity norm of the difference between the result we obtain using the method and the analytical solution calculated in the set R_{n+1} .

The results are grouped in the following table:

n	iterations' number	error
16	4	0,0012
32	4	$2.95 \cdot 10^{-4}$
64	4	$7.38 \cdot 10^{-5}$
128	4	$1.85 \cdot 10^{-5}$
256	4	$4.61 \cdot 10^{-6}$
512	4	$1.15 \cdot 10^{-6}$

We can clearly notice that, as we expected, with the raising values of n the distance between our solution and the actual solution decreases, as we can see in following graph:



We have also taken into account other three examples:

Example 1:

$$y'' - y^3 + x^6 - 2 = 0$$

$$y(0) = 0, \quad y(1) = 1,$$

Example 2:

$$y'' - \frac{(1+x+y)^3}{2} = 0$$

$$y(0) = 0, \quad y(1) = 1,$$

and Example 3:

$$y'' + y' - e^y = 0$$

$$y(0) = 0, \quad y(1) = 1,$$

in order to highlight the fact that our algorithm converges in a really small number of iterations. In the next table, indeed, we want to point out the number of iterations for the different examples of equations at the change of n .

n	number of iterations		
	Ex 1	Ex 2	Ex 3
4	5	5	4
8	5	5	4
16	4	5	4
32	4	5	4
64	4	5	4

We have also tackled the example (4.11) with the Newton's generalized method (3.29) using the relaxation factor $\omega = 1.3$ and we have obtained the following results:

n	iterations' number	error
16	20	0,0012
32	132	$2.95 \cdot 10^{-4}$
64	550	$7.38 \cdot 10^{-5}$
128	2141	$1.85 \cdot 10^{-5}$

The method converges to the same results of the Newton's method. The difference that we notice is that, even if the computational cost for iteration is smaller, the number of iterations that the Newton's generalized method requires is so much higher that the Newton's method is really more efficient.

Chapter 5

Conclusions

In this chapter we will briefly recap the topics studied in this thesis and the results obtained with our code in appendix A.

In Chapter 2 we have defined the Newton's method for non-linear equations and then the Newton's generalized method for systems of non-linear equations. We have studied and demonstrated some theorems for the convergence of the method. We have finally given some variations of the method.

In Chapter 3 we have studied the tridiagonal and diagonally dominant systems with some relatives theorems for the existence and uniqueness of the solution. We finally have studied the mildly non-linear systems and the theorem for its solution's existence and uniqueness.

In Chapter 4 we have applied the Newton's method studied in chapter 2 to a mildly non-linear boundary value problem. We have studied the conditions for it to have an analytical solution and the procedure to discretize it with the central difference formula. We have finally rewritten it as a mildly non-linear system which, if it satisfies condition (4.8), $Mh < 2$, has one and only one solution that can be found by the generalized Newton's method.

Lastly we have given an example of a mildly non-linear boundary value problem which has an analytical solution and we have shown how the choice of n is important for the outcome of the method: with an increase of n we have a decrease of the error.

In Appendix A we show the code, written in MatLab, that takes as input the features of the boundary value problem, a chosen $n \in \mathbb{N}$, the maximum number of iteration and the tolerance. The output of this code is 'error' if the system does not satisfies the condition (4.8), otherwise it's the solution of the system calculated in the set R_{n+1} .

Appendix A

Appendix

```
function[guess,niter, err]=metodoneutron(n, yi, yf, xi, xf, F, P, nmax,  
    ↪ toll)  
%prendo come input un'equazione differenziale del tipo y''+P(x)y'+F(x,y)  
    ↪ =0  
%con condizioni al bordo y(xi)=yi, y(xf)=yf  
  
%creo un reticolo di n punti equidistanti tra xi e xf  
h=(xf-xi)/(n);  
l=linspace(xi, xf, n+1);  
l=l(2: end -1);  
  
vettorey=linspace(yi,yf,n+1);  
guess=vettorey(2: end -1);  
  
%controllo la condizione per l'esistenza e unicita'  
w=abs(P(l));  
M=max(w);  
  
if M*h>2  
    error('Unique condition not satisfied')  
else  
    %uso il metodo delle differenze centrate per creare f=(f1,...,fn-1)  
    syms y [1,n-1]  
  
    f=differenzecentrate(y, l, h, n, yi, yf, F, P);  
  
    %ora che ho f posso utilizzare il metodo di newton con la guess  
    %iniziale e poi usare thomas per risolvere il sistema tridiagonale  
  
    J = jacobian(f, y);
```

```

        niter=0;

        for j=1:nmax
            x = guess;
            Jf = subs(J, y, guess);
            b = -subs(f, y, guess);
            guess = guess + (thomasbis(Jf, b))';
        niter=j;
        err=norm(guess-x, Inf);
        if err<toll, return, end
    end
end

```

where the functions called inside the method are the following:

```

function[f]=differenzecentrate(y, l, h, n, yi, yf, F, P)

syms f [1, n-1]

f(1) = (yi-2*y(1)+y(2))/h^2 + F(l(1), y(1)) +P(l(1))*((y(2)-yi)/(2*h));

for i=2:n-2
    f(i)=(y(i-1)-2*y(i)+y(i+1))/h^2+F(l(i), y(i))+P(l(i))*((y(i+1)-y(i-1)
        ↪ )/(2*h));
end

f(n-1) = (y(n-2)-2*y(n-1)+yf)/h^2 + F(l(n-1), y(n-1))+P(l(n-1))*(yf-y(n
    ↪ -2))/(2*h);

```

and

```

function[x]= thomasbis(J,f)
% Algoritmo di Thomas
% x=thomas(a,b,c,f) risolve il sistema T*x=f, dove T e' la matrice
    ↪ tridiagonale
% T=tridiag(b,a,c)=diag(a)+diag(b,-1)+diag(c,1)

a=diag(J);
m=length(a);
for i=1:m-1
    c(i)=J(i,i+1);
end
for i=2:m
    b(i-1)=J(i,i-1);

```

```

end

alpha=zeros(m,1); %U=diag(alpha)+diag(c,1)
beta=zeros(m-1,1); %L=diag(beta,-1)+eye(n)
y=zeros(m,1);
x=zeros(m,1);

alpha(1)=a(1);
y(1)=f(1);
for i=1:m-1
    beta(i)=b(i)/alpha(i);
    alpha(i+1)=a(i+1)-beta(i)*c(i);
    y(i+1)=f(i+1)-beta(i)*y(i);
end

x(m)=y(m)/alpha(m);
for i=m-1:-1:1
    x(i)=(y(i)-c(i)*x(i+1))/alpha(i);
end

return

```


Bibliography

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